Grounded C–G with Action Descriptions. For the grounded Action Description condition, the model sees symbolic descriptions for all examples and a *paired* description+image for the query:

Listing A.6: Minimal-Context Prompt

```
SYSTEM_PROMPT = """
    **Persona:** You are an expert visual reasoning system specializing in analyzing abstract
        patterns.

    **Objective:** Your primary goal is to solve a Bongard-style visual reasoning puzzle. You will
        be given abstract descriptions (action programs) used to generate synthetic images and a
        reference image. You must identify the underlying rule that distinguishes a "positive" set
        from a "negative" set.

    **Crucially, for the test sample, you will be given both its action program and the final
        rendered image.** Use this pair to understand the connection between the symbolic action
        programs and their visual generation. Apply this understanding to decipher the rule
        governing the positive and negative sets, for which you only have action programs. These
        action programs are the steps to draw the abstract shape on the canvas.

    **Your Task and Required Output:**
    Analyze the provided samples and respond *only* with a single JSON object. Do not include any
        additional text, explanations, or markdown formatting outside of the JSON structure.

    The JSON object must have the following keys:
    }
        "Analysis": "A brief analysis of the distinguishing characteristics you observed in the
            'pos' samples that make them distinct from 'neg' samples.",
        "Rule": "A clear and concise distinguishing rule that defines the positive samples.",
        "Test Image": "A summary of the test image's attributes, referencing the provided png
            image.",
        "Conclusion": "Your final categorization of the test image. The value must be either 'pos'
            or 'neg'."
    }
    """
```

Grounded C–G with Action Programs. For the grounded Action Program condition, the model receives LOGO-style programs for all examples and a program+image pair for the query, with an extended explanation of the program format:

Listing A.7: Minimal-Context Prompt

```
SYSTEM_PROMPT = """
    **Persona:** You are an expert visual reasoning system specializing in analyzing abstract
        patterns.

    **Objective:** Your primary goal is to solve a Bongard-style visual reasoning puzzle. You will
        be given abstract descriptions (action programs) used to generate synthetic images. You must
        identify the underlying rule that distinguishes a "positive" set from a "negative" set.

    **Input Format and Generative Context:**
    You will receive 13 sets of abstract figure descriptions enclosed in [...]. These descriptions
        are the **source code** (called "action programs") that programmatically generate the
        images. Understanding the generation process is key.

    * **The Generation Pipeline:**
        1. **Stage 1: Base Shape Definition:** Shapes are first defined with basic geometry using
            degrees for angles.
            * `set of base actions`: A list of primitives like `line_LENGTH` or `arc_RADIUS_ANGLE`.
            * `turn angles`: A sequence of turns like `L90.0--R45.0...`, specified in degrees.
        2. **Stage 2: Object Creation (`BasicAction`):** The system reads the base geometry and
            creates programmable `BasicAction` objects. At this stage, a visual stroke style
            (`line_type` like "zigzag" or "triangle") is added.
        3. **Stage 3: Final Serialization (The Input You See):** The `BasicAction` objects are
            serialized into the final action program strings. All values are **normalized** into a
            [0, 1] range.
            * **Line String:** `line_TYPE_LENGTH-TURNANGLE`
            * **Arc String:** `arc_TYPE_ARCANGLE_ARCRADIUS-TURNANGLE`

```

```
    * **Program Structure Hierarchy:**
        The final action program is a nested list: `BongardProblem` -> `BongardImage` ->
            `OneStrokeShape` -> `BasicAction`.

    * **Data Sets You Will Analyze:**
        1. **Positive Samples (`pos`):** Action programs that all conform to a common abstract rule.
        2. **Negative Samples (`neg`):** Action programs that do not conform to this rule.
        3. **Test Sample:** An action program to be categorized, accompanied by its rendered image
            to provide visual context.

    * **Crucial Interpretive Note:** While the action programs detail *how* a figure is constructed,
         your task is to reason about the final, abstract geometric properties of the resulting
         figure. The construction method is just one of several possible ways to draw the same shape.

    **Your Task and Required Output:**
    Analyze the provided samples and respond *only* with a single JSON object. Do not include any
         additional text, explanations, or markdown formatting outside of the JSON structure.

    The JSON object must have the following keys:
    }
        "Analysis": "A brief analysis of the distinguishing characteristics you observed in the
            'pos' samples that make them distinct from 'neg' samples.",
        "Rule": "A clear and concise distinguishing rule that defines the positive samples.",
        "Test Image": "A summary of the test image's attributes, referencing the provided png
            image.",
        "Conclusion": "Your final categorization of the test image. The value must be either 'pos'
            or 'neg'."
    }
    """
```

D.7 User Prompt

We used the same user prompt for all our experiments.

Listing A.8: Natural Language Procedure Prompt

```
USER_PROMPT = """
POSITIVE SET (6 descriptions):
[Description 1]
[Description 2]
[Description 3]
[Description 4]
[Description 5]
[Description 6]

NEGATIVE SET (6 descriptions):
[Description 7]
[Description 8]
[Description 9]
[Description 10]
[Description 11]
[Description 12]

QUERY (1 description):
[Description 13]

Classify the query as 'positive' or 'negative'. Respond with JSON
only.
"""
```

E Extended Results Tables

E.1 Complete Results by Model and Condition

Table A.3 reports full C–G performance on the Bongard-LOGO benchmark across all 12 models and experimental conditions. For each model, we include Action Description, Action Program, and minimal-context Base (Ablation) conditions, broken down by Basic Shapes (BD), Free-form (FF), Human-designed Combined (HD Comb), and Human-designed Novel (HD Novel).

Table A.3: C-G paradigm performance on Bongard-LOGO. Results show accuracy (%) for Action Desc. and Action Prog. across problem categories with concept (C) and without concept (NC) guidance. BD = Basic Shapes, FF = Free-form, HD = Human-designed. Action Desc. = Action description and Action Prog. = Action program.

Model	Experiment	BD		FF		HD Comb		HD Novel	
		C	NC	C	NC	C	NC	C	NC
DeepSeek-R1									
8b	Action Desc.	61.8	65.6	61.6	63	50.6	56.2	52.2	54.6
	Action Prog.	56	57.52	56	56.2	53.4	54.91	55.2	53.8
	Baseline		53.0		51.4		48.8		53.0
14b	Action Desc.	70.6	70.4	75.4	74	58.6	57	58.6	57.6
	Action Prog.	69.6	68.8	75.6	77.8	56.8	57.8	59	57.6
	Baseline		71.8		74.6		59.2		58.6
32b	Action Desc.	69.2	64.6	78.2	82.8	55.6	59.6	59.6	58.6
	Action Prog.	67.4	65.2	81.2	78.6	59.4	57.8	59.2	64.2
	Baseline		66.2		74.6		55.8		58.2
Gemma3									
12b	Action Desc.	65	65.06	72.8	69.2	58.6	56.6	61	65.8
	Action Prog.	62.8	64	71.4	70	63	65.4	62.4	63.6
	Baseline		66.6		71.2		60.8		62.2
27b	Action Desc.	70.2	65.6	79.4	81	64.2	57.4	70	62.6
	Action Prog.	69.4	67	87.4	83.2	64.4	66	64.4	66.6
	Baseline		67.6		82.8		63.0		66.8
Magistral:24b	Action Desc.	73.8	75.8	85.8	84.8	60.6	60.4	66	63.2
	Action Prog.	72.8	70.6	81.6	80.4	57.8	60	67.2	58.2
	Baseline		72.6		82.2		59.4		64.2
Phi4:14b	Action Desc.	68.8	75.2	91.8	90.2	58.8	55	59.6	58
	Action Prog.	71.6	70.8	92	89.8	60.2	62.2	64.8	62.4
	Baseline		70.2		97.0		61.2		61.4
Phi4-Reasoning:14b	Action Desc.	73.4	79.4	97.4	87.8	57.2	51.8	60.4	58.8
	Action Prog.	75.6	72.6	94.2	96.2	60	59.4	62.8	63.2
	Baseline		70.2		97.0		55.6		59.0
Qwen2.5:32b	Action Desc.	71.8	73.8	75	80.8	60.8	60.2	64.2	61.4
	Action Prog.	74.4	73.6	78.2	76.2	63.6	62.2	65	63
	Baseline		71.4		75.8		63.4		60.8
Qwen2.5vl:32b	Action Desc.	58.4	68.6	70.8	72.8	59	59.6	58.6	59.4
	Action Prog.	67.2	66.8	73	72.4	62.6	60.2	61	61.2
	Baseline		60.2		65.6		60.8		56.2
Qwen 3									
30b	Action Desc.	74.4	75.2	84	78.2	60.6	56.8	62.6	63.2
	Action Prog.	75.6	72.6	79.2	77.4	59.4	59.4	60.8	61.4
	Baseline		65.4		74.0		58.6		61.4
32b	Action Desc.	78.4	84.6	81.6	86.8	61.6	58.8	60.6	64.6
	Action Prog.	77.6	76.4	82.2	78.8	61.8	61	62.4	61.4
	Baseline		77.4		81.8		59.4		63.4